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Toward comprehensive cellular characterization of H&E slides

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ABSTRACT

Cell detection, segmentation, and classification are essential for analyzing tumor microenvironments (TME) on hematoxylin and eosin (H&E) slides. Existing methods suffer from poor performance on understudied cell types (rare or not present in public datasets) and limited cross-domain generalization. To address these shortcomings, we introduce HistoPLUS, a state-of-the-art model for cell analysis, trained on a novel curated pan-cancer dataset of 108,722 nuclei covering 13 cell types. In external validation across 4 independent cohorts, HistoPLUS outperforms current state-of-the-art models in detection quality by 5.2% and overall F1 classification score by 23.7%, while using 5× fewer parameters. In addition, we show that HistoPLUS robustly transfers to two oncology indications unseen during training and allows interpretable biomarker discovery in downstream tasks, outperforming clinical baselines and prior deep learning methods. To support broader TME biomarker research, we release the model weights and inference code.

Introduction

Cell detection, segmentation, and classification in histological images form the foundation of modern computational pathology, enabling quantitative analysis of the tumor microenvironment (TME) for applications in patient stratification[1,2] and personalized medicine.[3–5] Beyond cell identification, these techniques reveal spatial relationships between tumor cells, immune cells, and stromal compartments that correlate with disease outcomes[6,7] and treatment responses.[8,9] The spatial organization and density of specific cell populations—such as tumor-associated neutrophils (linked to immunosuppression[10]) or cancer-associated fibroblasts (implicated in therapy resistance[11–13])—have become increasingly important in clinical research and trials.[14,15] Whereas immunohistochemical methods provide targeted molecular information, hematoxylin and eosin (H&E) staining remains predominant in clinical practice due to its accessibility, cost-effectiveness, and rich morphological detail.[16]

However, extracting comprehensive cellular information from H&E images presents substantial computational challenges. Cells in tissue sections exhibit marked heterogeneity in morphology, size, and staining characteristics, while often being densely packed with indistinct boundaries. The identification of clinically relevant immune cells (e.g., lymphocytes, neutrophils, eosinophils, and plasmocytes) and stromal cell subsets (e.g., fibroblasts, smooth muscle cells) requires extensive annotated training data from expert pathologists. Yet, to the best of our knowledge, current public annotated datasets have critical shortcomings: many datasets provide segmentation masks without classification labels,[17–19] limiting utility for TME studies; others offer only bounding boxes,[20] precluding morphometric analysis. Even datasets with cell-type annotations have limited class granularity,[21–23] lack rare cell representation,[21] or focus on a limited set of cancer types.[24] These constraints, coupled with insufficient sample sizes,[23,25] have hindered the development of robust, generalizable models across diverse tissues and diseases.

Recent pathology foundation models (PFMs) pre-trained via self-supervision[26–28] promise to mitigate these limitations by leveraging unlabeled histology data to learn transferable representations, reducing reliance on scarce expert annotations. Models like UNI,[29] Phikon,[30,31] and H-Optimus-0[32] have advanced patch classification and molecular prediction,[33] but their applicability to cell detection, segmentation, and classification—particularly for understudied subtypes i.e., non-lymphoid immune cells (eosinophils, neutrophils, and macrophages), plasmocytes, stromal cells (smooth muscle, endothelial), and rare events (mitotic figures, apoptotic bodies)—has not been thoroughly investigated. Despite evaluations of Hibou-Large,[34] UNI, Virchow, and GigaPath[35] on cell-level tasks, no benchmark compares PFMs performance rigorously and exhaustively

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for simultaneous cell detection, segmentation, and classification tasks. This gap is critical, as without systematic benchmarking of PFMs, researchers cannot identify which models maximize performance for these tasks—leaving potential performance gains unrealized despite their importance for quantifying the TME.

To address these gaps, we present:

1) A pan-tumor dataset with granular labels for 13 cell types across 6 cancer indications, obtained using an active-learning-based annotation pipeline that maximizes label quality and diversity.
2) A PFM-integrated model for pan-cancer cell detection, segmentation. and classification in H&E, leveraging self-supervised distilled features[31,32,36] within a CellViT architecture that achieves state-of-the-art performances on all nuclei identification, including understudied types.

Our model demonstrates superior performance both in cross-validation and on external validation. It paves the way for new analyses of spatial tumor–immune interactions. To facilitate community adoption and further progress in computational pathology, we release our pre-trained CellViT model. This contribution provides the research community with a state-of-the-art cell detection, segmentation, and classification model on H&E slides. It aims to accelerate biomarker discovery, ultimately enhancing our ability to translate histological insights into clinically relevant applications.

Methods

Data collection for training set

To generate our dataset, we relied on TCGA datasets of colon adenocarcinoma (COAD), lung adenocarcinoma (LUAD), lung squamous cell carcinoma (LUSC), mesothelioma (MESO), bladder urothelial carcinoma (BLCA), ovarian serous cystadenocarcinoma (OV), pancreatic adenocarcinoma (PAAD), and breast invasive carcinoma (BRCA). As a first step, whole-slide images (WSI) are divided into smaller, non-overlapping areas, referred to as tiles. Each tile represents a tissue portion of 112 µm × 112 µm either at 40× when available (n = 1322), or at 20× (n = 93). We leveraged pretrained foundation models[31] to extract tile-level features. We selected an equal number of tiles for each indication, using our active learning pipeline to favorize diverse and informative tiles within cohorts. Selected tiles were then uploaded to Cytomine,[37] a web-based interface allowing board-certified pathologists to point nuclei centroids, and select the corresponding cell type among a predefined ontology. A "Cell with unknown type" class was also available if pathologists were not fully confident on their guess. Annotations were then pulled from Cytomine for preprocessing.

Data collection for validation sets

In parallel with collecting training annotations, we collected annotations for validating our final model on external cohorts, relying on MOSAIC datasets. Detailed methods associated with the MOSAIC data are presented in Dong et al.[38] Within available datasets, 246 slides were selected randomly and N = 530 tiles within these slides are selected for annotation. This selection is done by random selection (N = 340), by maximizing the probability of presence of eosinophils (N = 99) and by maximizing the presence of neutrophils (N = 91). Presence of neutrophils and eosinophils is assessed based on the predictions of an ensemble of three multi-layered-perceptron (MLP) trained to predict presence of nuclei types at the tile level. These MLPs consist of a single-layer MLP with an input layer of size 2048 (taking as input MoCo-WideResNet[39] extracted features) and an output of size 1, with a Sigmoid activation, predicting the presence of at least one cell of the corresponding cell type within the tile. These MLPs were pretrained on 3 datasets (TCGA LUAD, LUSC, and COAD), where 636 tiles were annotated (307 tiles were annotated positive for the presence of eosinophils and 329 for the presence of neutrophils). We also randomly sample the same number of negative examples from our dataset to obtain a balanced representation in the training set of each predictor. This methodology ensures a decent representation of important understudied nuclei types (Supplementary Fig. 2). These positive and negative examples were selected from patients distinct from patients used in HistoTRAIN training set.

Building a diverse dataset with active learning

Using the tiles presented in the Data collection paragraph, we chose regions of interest using: (A) clustering methods to ensure tissue diversity and (B) active learning-based strategies to maximize the representation of areas with rare cells or ambiguous cells. As the inference of large cell detection, segmentation, and classification is computationally expensive, we used tile-level classifiers performances (simple predictors estimating the presence/absence of a given cell type within a tile) as proxies for estimating the classification performance of cell detection, segmentation, and classification models. It allowed us to leverage active learning strategies (iteratively refine classifiers thanks to newly collected annotations) without the compute and time cost of fitting large cell detection, segmentation, and classification models at each iteration. We defined three different active learning methods to select tiles within candidate datasets; our goal was to not only generate tiles for annotation but also to evaluate the performance of tile selection strategies and the added value of rare cell enrichment and uncertainty sampling.

The first method, used as baseline, consisted of feature-based clustering: given a pool of S slides, we performed K-means clustering of the tiles for each slide based on their features extracted using Phikon.[31] We sampled randomly one tile from each cluster, yielding K*S tiles for the dataset. We then randomly sampled the desired number of tiles N_0 from this pool. For our experiments, we set K = 20 as suggested in the recent literature,[40] to ensure homogeneous, diverse tile clusters across the slide.

Our second method was designed to enrich our dataset in under-represented cell types, such as: apoptotic bodies, mitotic figures, eosinophils, endothelial cells, epithelial cells, macrophages, and neutrophils. For every phenotype, we trained lightweight MLPs that map either Phikon or MoCo-WideResNet[39] features to a probability of presence, using annotated tiles from TCGA COAD, LUAD, and LUSC cohorts and all tiles annotated from the previous iterations (resulting in 341 tiles for the first iteration and 1782 tiles after all iterations). We performed cross-validation on cohort-stratified folds with nested inner-folds, features are z-scored with a conditional scaler (global, slide-wise, or cohort-wise) to cancel slide/cohort biases. Twelve models (2 network depths × 2 feature sets × 3 scalers) were compared per cell type and the best one, estimated using cross-validation balanced accuracy, was refitted on the entire dataset for inference. We scored all available tiles and kept the top two highest scored tiles, per slide and per cell type. The union of these high-confidence tiles were merged with the diversity pool before the final sampling step (random sampling from the pool), ensuring that rarer cell types are oversampled, whereas the slide-level weighting keeps the overall distribution balanced.

To further prioritize informative examples, we computed *epistemic uncertainty* on the rare-cell predictions in our third arm. Each phenotype model was run as an ensemble; from the per-tile logits we derived entropy, variation ratio, and BALD scores.[41] Among tiles whose predicted probability were above 0.4, the ones whose BALD score (high uncertainty) exceeded the 90th percentile on their slide were retained. For those uncertain tiles, we performed K-means on each slide (between 5 and 20 clusters depending on remaining tile count) and took one representative per cluster. This yielded a compact uncertainty-aware pool. During the final draw, we sampled 50 tiles per indication, proportional to the number of uncertain tiles a slide contributes but clipping slide weights to the 0.5–2.0 range to avoid over- or under-representation.

This resulted in a dataset that: (i) covered the morphological space of every slide, (ii) deliberately enriched rare phenotypes, and (iii) focused annotation effort on regions where the current models disagree most, accelerating both cell-type and segmentation learning in later active-learning rounds.

Cell boundary extension with Nuclick

Handmade segmentations are time-consuming to perform and highly subject to pathologist inter-variability, which prevents building a large, high-quality set of annotations for model training and testing, across multiple indications. In order to speed up and robustify the annotation collection process, we asked pathologists to solely annotate the centroid of each nucleus with a point annotation along with the corresponding cell type, and processed these pre-annotations using Nuclick[39] to retrieve nuclei segmentations from pathologist point annotations. Nuclick is a CNN-based prompted segmentation network which takes as input a map of annotated nuclei centroids within a tile and returns the corresponding nuclei segmentations as an instance map. We used the vanilla Nuclick U-Net architecture as described in the official Github repository (https://github.com/mostafajahanifar/nuclick_torch/), and trained our model on ConSEP[22] and PCNS[42] datasets, which regroups annotated data from 14 cancer indications. For each annotated nucleus in the training set, we extracted a 128 × 128 patch centered on the cell, and generated inclusion and exclusion maps using segmentation centroids (to which slight Gaussian noise was added) to mimic user-generated points. We followed the official train/test split for ConSEP and used all extracted patches from PCNS for training. This resulted in 46,000 training and 8600 validation patches. Training was performed for 60 epochs with a learning rate of 0.001, a batch size of 64 and weighted BCE Dice loss, as described in the original article, and we monitored validation performance using Dice score. Our best model achieved a Dice score of 0.874 in validation after 35 epochs, and was used for inference on collected pathologist annotations. Nuclick segmentations are then used as ground-truth (GT) annotations for model training.

Consensus-based annotation framework for robust external validation

Creating reliable GT for nuclei segmentation on H&E images is difficult due to high inter-annotator variability among expert pathologists, primarily in cell type attribution and annotation exhaustivity. To mitigate this, we collected annotations from three pathologists for each tile and used a consensus framework to generate curated and reliable external validation sets.

Our consensus framework operates in four sequential steps. First, 2–3 pathologists independently annotate nuclei centroids with point annotations on a given set of tiles (number of total annotations before consensus = 212,992 nuclei). Second, we infer Nuclick to expand the points to nuclei segmentations. Third, we establish correspondence between annotations from different pathologists by matching segmentations with an Intersection over Union (IoU) above 0.4, retaining only nuclei identified by at least two pathologists to ensure reliability. Finally, for each matched nucleus, we compute a consensus centroid by averaging all corresponding centroid annotations. The cell type is determined through majority voting. The final consensus segmentations are generated by applying NuClick to the computed consensus centroids, yielding high-quality nuclear boundaries paired with robust cell type classifications. We applied this methodology to derive the whole HistoVAL dataset, a validation dataset which contains 530 regions of size 112 µm × 112 µm and a total of N = 69,108 consensus-driven nuclei.

Vision Transformer

The Vision Transformer (ViT) processes 2D images by dividing the input into a sequence of flattened patches, which are linearly projected into a D-dimensional embedding space and combined with learnable positional embeddings to retain spatial information. A prepended [CLS] token serves as the global image representation for classification. The Transformer encoder consists of alternating multi-head self-attention[43] and MLP blocks, each preceded by a LayerNorm[44] operation and followed by residual connections. The MLP employs two layers with Gaussian Error Linear Unit activation. Unlike convolutional neural networks, ViT relies primarily on self-attention to learn spatial relationships, with limited image-specific inductive biases.

CellViT

Model description

CellViT[45] adapts the UNETR architecture[46] for 2D histopathology images by integrating a ViT[47] encoder with a multi-task decoder structure inspired by HoVerNet. The model employs three distinct output branches: (1) a nuclei prediction (NP) branch for binary segmentation of nuclei boundaries, (2) a horizontal-vertical (HV) branch generating normalized distance maps to nuclei centers, and (3) a nuclei type branch for instance-aware classification. The ViT encoder processes image patches as token sequences with learnable positional embeddings, whereas skip connections fuse multi-scale features from five encoder stages into the decoder. Unlike traditional U-Nets, CellViT's decoder uses isolated upsampling pathways for each branch, maintaining input resolution in the output. We selected different layers based on the size of the ViT encoder.

Loss

CellViT employs a multi-task loss combining weighted branch-specific objectives: (1) a NP loss using cross-entropy and Dice to segment nuclei boundaries, (2) a HV loss with mean squared error on distance maps and their gradients for spatial localization, and (3) a nuclei type loss with Focal Tversky to address class imbalance in fine-grained classification. The Focal Tversky loss prioritizes under-represented nuclei classes through adjustable hyperparameters, whereas the total loss balances contributions via weights. We used the exact loss weights as presented in the original CellViT implementation (Table A.3).

Postprocessing

To generate instance-aware segmentation from CellViT's multi-branch outputs, we use the original two-stage approach, presented in HoVerNet. For individual patches, nuclei separation is achieved by: (1) computing gradients of the HV distance maps to detect boundary transitions, (2) applying Sobel edge detection to highlight nuclei contours, and (3) using marker-controlled watershed to resolve overlapping instances. Nuclei class assignments are derived via majority voting on the NT prediction map within each segmented region.

Handling multiple image sizes

CellViT processes input images through a ViT backbone pretrained at a fixed resolution of 224 × 224, with positional encodings dynamically interpolated to accommodate various input sizes. The encoder yields feature maps of spatial dimension 16 × 16 (for patch size 14) and 14 × 14 (for patch size 16), derived from the ratio of input resolution to patch size. During decoding, these feature maps are progressively upsampled ($\times 2^4$) to 256 × 256 (for patch size 14) or 224 × 224 (for patch size 16). For the 256 × 256 outputs, a final downscaling to 224 × 224 ensures consistency in loss computation and metric evaluation. This design preserves the positional information of the patches across scales while maintaining flexibility for multi-scale inputs.

Pathology foundation models

Self-supervised learning

Self-supervised learning (SSL) enables models like ViTs to learn robust feature representations from unlabeled data by solving pretext tasks, eliminating the need for costly manual annotations. In computational pathology, SSL frameworks like iBOT,[26] DINO, and DINOv2[27] are widely used to pretrain foundation models by learning representations from augmented or masked image patches. In this study, the evaluated foundation models were pretrained using these SSL approaches: UNI, UNI2, Virchow, and Virchow2G were trained using DINOv2, whereas Phikon and PhikonV2 were trained using iBOT and DINOv2, respectively.

Knowledge distillation

Knowledge distillation in SSL refers to the process of transferring knowledge from a large, pretrained teacher model to a smaller, more efficient

student model without relying on labeled data. In SSL-based PFMs, this can involve matching global image-level representations and local patch-level features across augmented views. H0-mini is a compact distilled model derived from H-Optimus-0 that combines DINO-style class-token distillation with iBOT-style patch-level supervision, providing an efficient foundation model for downstream histopathology tasks.

Evaluation metrics for detection, segmentation, and classification

Detection and segmentation metrics

To rigorously evaluate the segmentation performance of our model, we employed the panoptic quality (PQ) metric, which provides a comprehensive assessment by jointly considering detection accuracy and segmentation precision. The PQ score decomposes into two interpretable components:

1. Detection quality (DQ), analogous to the F1-score, which quantifies the correctness of instance detection,
2. Segmentation quality (SQ), defined as the average IoU of matched segments.

Instance matching was performed using the Munkres (Hungarian) algorithm,[48,49] an optimal assignment method that minimizes the total Euclidean distance between GT and predicted centroids, ensuring one-to-one correspondence. Following established conventions,[50] we consider a segment pair (y, y_hat) as a true positive (TP) only if their IoU exceeds 0.5, ensuring unique matching between predicted and GT instances. Unmatched predictions and GT segments are classified as false positives (FPs) and false negatives (FNs), respectively. We report the binary PQ, treating all nuclei as a single class. This approach aligns with recent benchmarks.[22,45]

Classification metrics

Classification of nuclear type is performed on the instances extracted from either instance segmentation or detection. Thus, the evaluation of nuclear type classification must account for both detection and classification accuracy. For each nuclear type t, the detection task (d) partitions GT and predicted instances into correctly detected instances (TP_d), missed GT instances (FN_d), and over-detected predictions (FP_d). The classification task (c) further divides TP_d into correctly classified instances of type t (TP_c), correctly classified instances of other types (TN_c), misclassified instances of type t (FP_c), and misclassified instances of other types (FN_c).

Classification metrics in the case of multi-label annotation

When the consensus yields several GT cell types, the previous approach cannot be used. Instead, for each nuclear type t, FP_d is classically defined as the number of unpaired predicted cells whose prediction is t, whereas FN_d is defined as the number of unpaired true cells whose GT labels list contains t. For classification, we further define TP_c if t is in the GT labels list when the prediction is t, TN_c if t is not in the GT labels list and the prediction is not t, FP_c if t is not in the GT labels list and the prediction is t and FN_c if both t is in the GT labels list, the prediction is t and the prediction is not in GT labels list. Confidence intervals at 95% confidence level were obtained by bootstrapping experiment results with 1000 repeats. All tests were two-tailed, and p-values <0.05 were considered statistically significant.

External datasets

PanNuke

The PanNuke dataset[21] consists of 30,000 nuclei-annotated tiles split into 3 folds on H&E images, coming from 19 different tissue types (adrenal, bile duct, bladder, breast, colon, cervix, esophagus, head and neck, kidney, liver, lung, ovarian, pancreatic, prostate, skin, testis, stomach, thyroid, and uterus). Cells were annotated by trained pathologists on five different classes: neoplastic, non-neoplastic epithelial, inflammatory, connective, and dead. Each image has a dimension of 256 × 256 pixels at a 40× magnification.

Lizard

The Lizard dataset[24] consists of 6 annotated datasets (DigestPath, CRAG, GlaS, PanNuke, CoNSeP, and TCGA) for 495,179 annotated nuclei, coming from colon H&E slides. Cells were annotated by trained pathologists on six different classes: epithelial, connective, lymphocyte, plasma, neutrophil, and eosinophil. It is composed of 291 image regions with an average size of 1016 × 917 pixels at a 20× magnification. We further preprocessed the raw images to create a preprocessed dataset consisting of images of size 224 × 224 pixels.

Benchmarking of CellViT on HistoTRAIN

All models were trained using 3-fold cross-validation using the folds presented in Gamper et al.[21] on 4 Tesla T4 GPUs, with DeepSpeed Stage 0 (Data Parallelism) in mixed precision (FP16) and CPU offloading. Input images (40× magnification) of size 448 × 448 were directly fed to the CellViT model (see the Handling multiple image sizes paragraph for more details). We used Adam as an optimizer with a global batch size of 16 (4 per GPU), with an initial learning rate of 1e-5, weight decay of 1e-5 and betas (0.9, 0.99). We did not use any learning rate scheduler, as we observed in our experiments that a low constant learning rate offered the lowest and steadiest decrease in validation loss. The models were trained for 150 epochs with validation every 10 epochs to reproduce the training setup of Gamper et al.[21] We created skip connections that fuse features from encoder layers 3, 6, 9, and 12 (ViT-Base), layers 6, 12, 18, and 24 (ViT-Large), and layers 8, 16, 24, and 32 (ViT-Huge), ensuring consistent multi-scale fusion across model sizes. We used all the data augmentation parameters presented in the original CellViT implementation (Table A.2)[45] and the same over-sampling strategy to account for the cell distribution imbalance (paragraph 4.4 with gamma = 0.85). For full reproducibility, all experiments used a fixed random seed (111) for deterministic initialization and data shuffling.

External validation on HistoVAL

To further assess the generalization ability of the cross-validated models on HistoTRAIN, the model from the last fold and last epoch was selected for external validation on HistoVAL.

Univariate association with analysis for FGFR3 mutation prediction

To identify cell-level features associated with FGFR3 mutation status, we extracted 72 histomorphometric covariates from HistoPLUS outputs corresponding to the seven best-predicted cell types: cancer cells, lymphocytes, fibroblasts, neutrophils, eosinophils, plasmocytes, and smooth muscle cells. For each covariate, we fitted a univariate logistic regression model with an intercept on the standardized feature values using 5-fold cross-validation with 3 repeats. We ranked models by mean cross-validated AUC and selected the top 10 performing covariates for further analysis. For each selected covariate, we computed odds ratios with 95% confidence intervals derived from the t-distribution, along with associated p-values. To account for multiple hypothesis testing, p-values were corrected using the Benjamini–Hochberg false-discovery rate (FDR) method. All statistical analyses were performed using scikit-learn 1.6.1 and statsmodels 0.14.6.

Results

Pan-cancer nuclei dataset enriched in understudied types enables granular study of TME

We introduce HistoTRAIN, a pan-cancer nuclei dataset designed to address critical gaps in cellular diversity, annotation quality, and clinical applicability. The dataset comprises 108,722 nuclei segmentations from 739 H&E WSI, across 6 cancer types (BLCA, COAD, LUAD, LUSC, MESO, and PAAD) and covering 13 distinct cell types. Dataset construction followed

a two-step process: expert pathologists first provided point annotations of nuclei centroids with corresponding cell-type labels, after which NuClick[51] —a deep learning model trained to infer nuclei boundaries from point annotations—was used to generate precise segmentations. This approach substantially reduces inter-annotator variability in boundary delineation while accelerating large-scale annotation.

To systematically enrich rare and clinically relevant cell populations, we implemented an active learning pipeline (Fig. 1a) that iteratively selects

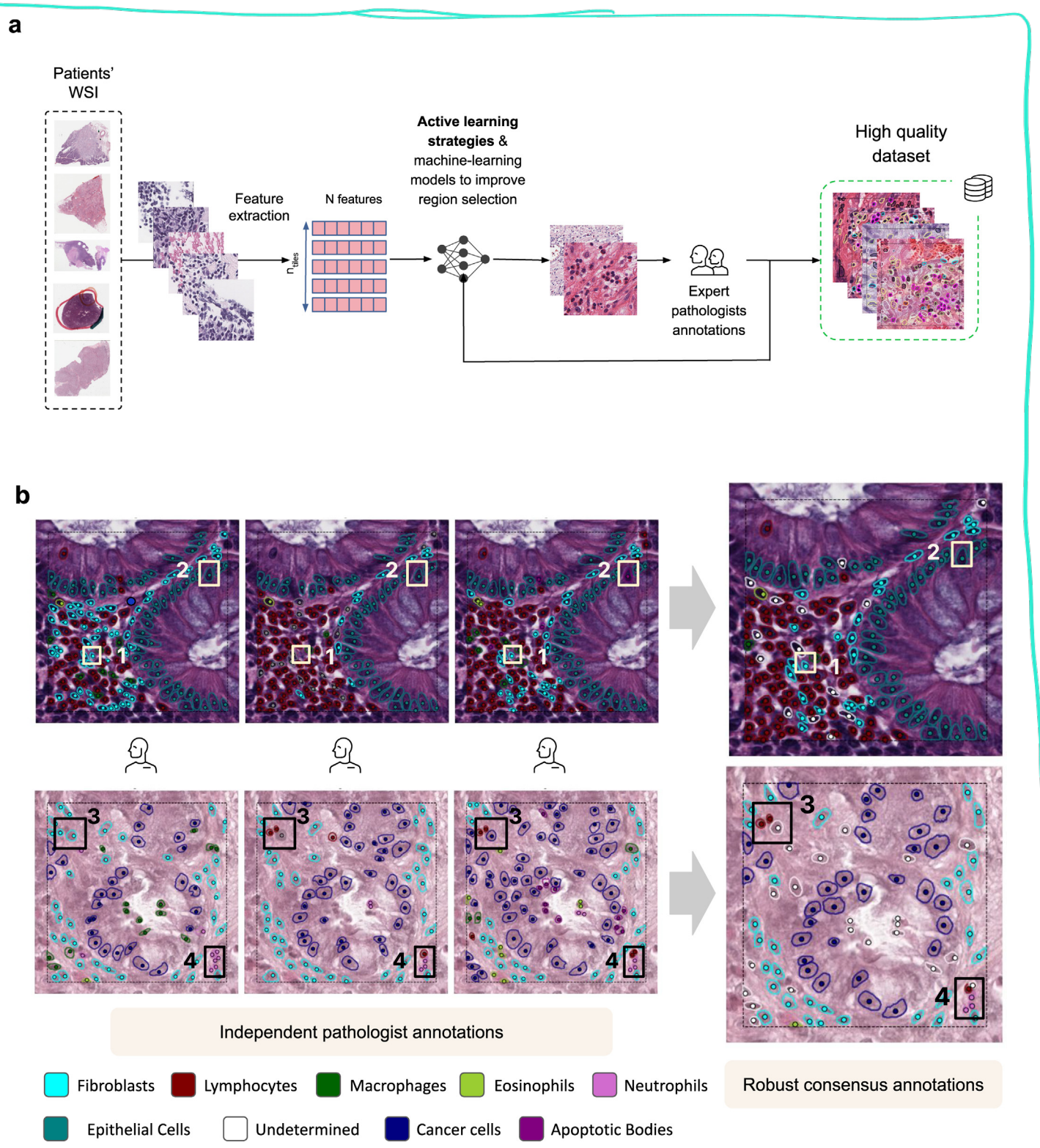


Fig. 1. Overview of our training dataset. (a) Candidate whole-slide images (WSIs) are divided into tiles, and features are extracted from each tile using Phikon, a pathology foundation model. To prioritize informative and diverse tiles, we apply K-means clustering based on features as well as multiple active learning strategies: rare cell type enrichment using cell type predictors (MLPs) and uncertainty estimation. The set of tiles is submitted to a pool of expert pathologists for cell type annotation. (b) Example of consensus annotations derived from regions annotated independently by three expert-pathologists. Disagreement may arise from distinct nuclei chosen to be annotated vs. skipped,[1] distinct cell types being attributed,[2] or a mix of both of these reasons.[3,4]

tissue regions with low model confidence and a high likelihood of containing understudied cell types (see Methods). Compared with existing public datasets (Table 1), this strategy increased the representation of non-lymphoid immune cells (eosinophils, neutrophils, and macrophages), plasmocytes, stromal cells (smooth muscle, endothelial), and rare events such as mitotic figures and apoptotic bodies. Non-malignant epithelial cells and red blood cells were also included to preserve morphological context. Cell-type distributions across cancer indications are detailed in Supplementary Fig. 1.

Whereas this strategy improves the diversity and scale of the training data, rigorous evaluation requires independently curated GT with controlled annotation variability. Deriving a reliable GT for cell detection, segmentation, and classification tasks on H&E tasks is challenging due to substantial inter-annotator variability, particularly in cell-type assignment and annotation exhaustivity.[52] To address these challenges, we randomly assigned three out of nine expert pathologists to annotate each tile and used a consensus framework to create curated, reliable external validation sets (Fig. 1b). We applied this methodology to derive HistoVAL (see Methods), a validation dataset which contains 530 regions of size 112 µm × 112 µm acquired at 40× magnification, selected from 246 distinct slides, covering the same 13 cell types. HistoVAL represents a total of N = 69,108 consensus-driven nuclei and covers external cohorts for four cancer types present in the training set to assess external cohort transferability and two unseen cancer types (ovarian, breast) to further assess transferability.

3.2

Pathology foundation models improve cell classification

To evaluate the impact of pathology-specific pretraining on cell detection, segmentation, and classification, we integrated PFMs as encoders within the fixed CellViT[45] architecture and compared them with the general-purpose Segment Anything Model[53] (SAM) using 3-fold cross-validation stratified by cell type on HistoTRAIN. To further evaluate the performance of each model to generalize, we performed external validation on HistoVAL. For cell detection and segmentation, we used the DQ and SQ metrics. For cell classification, we computed F1 scores for each cell type (see Evaluation metrics for detection, segmentation and classification in Methods).

To compare equal-sized encoders, we evaluated PFMs built on the ViT[47] base architecture against the general purpose encoder SAM-Base[53] (SAM—B). Domain-specific pretraining consistently improved performance over general-purpose models in cross-validation (Supplementary Tables 1, 2). These findings are also confirmed in external validation (n = 530): Phikon,[31] Hibou-B,[34] and H0-mini[36] showed a respective relative improvement of 16.5%, 17.2%, and 36.5% in their average F1 score compared to their counterpart SAM-B. For example, H0-mini achieved an F1 score of 0.583 [0.544–0.619] on tumor cells compared to SAM-B's 0.456 [0.414–0.493]. For understudied cell types, the improvements were even more substantial and statistically significant, showing H0-mini achieved F1 scores of 0.483 [0.433–0.529] vs. 0.269 [0.219–0.322] for plasmocytes ($p < 0.001$) and 0.403 [0.282–0.515] vs. 0.186 [0.110–0.257] for smooth muscle cells ($p < 0.001$) (Fig. 2a; Supplementary Table 4). Similar results are found as well for large encoders: Virchow2[54] and UNI2[29] showed a respective improvement of 17.4% and 26.7% compared to SAM-H.[53]

Despite considerable increases in model complexity, larger architectures showed limited performance gains while incurring significant computational costs. ViT-Huge models (UNI2[29]; Virchow 2[54]; 632M parameters) achieved only marginal PQ improvements over the compact H0-mini[36] model (86M parameters) in cross-validation (Supplementary Table 1). External validation revealed similarly modest or no improvements, with UNI2[29] and Virchow 2[54] both achieving DQ scores of 0.753 [0.742–0.763], compared to H0-mini[36] 0.753 [0.742–0.763] (Supplementary Table 3). For cell classification, Virchow 2[54] and UNI2[29] achieved a F1 score on cancer cells of 0.579 [0.544–0.613] and 0.603 [0.570–0.634], respectively, in-line with H0-mini's 0.583 [0.544–0.619] (Supplementary Table 4). Similar patterns were observed across individual cell types, including rare classes where larger models failed to substantially outperform H0-mini[36] (Supplementary Table 4). These marginal performance gains came at significant computational cost: 5.1× more parameters and 2.1× times longer inference time (Fig. 3a; Supplementary Table 5). Besides, SQ remained consistent across all model sizes ($\Delta SQ < 1.5$), confirming that the primary benefit of domain-specific pretraining lies in classification accuracy rather than boundary delineation.

Taken together, these results motivated the selection of HistoPLUS, a compact yet high-performing configuration combining the CellViT[45] architecture with the H0-mini[36] backbone, as our reference model (Fig. 2b). HistoPLUS offers an optimal balance between accuracy and efficiency, with 5× fewer parameters than CellViT SAM-H[45] while matching its DQ and exceeding its classification performance, particularly for rare and clinically relevant cell types (Figs. 2c–3b). This balance enables accurate, fine-grained TME profiling at substantially reduced computational cost, making HistoPLUS a practical choice for large-scale deployment.

3.3

HistoPLUS demonstrates robust performance on unseen oncology indications

To assess the generalizability of HistoPLUS beyond the indications present in the training set (Supplementary Tables 6 and 7 for performances per indication), we included in our validation datasets samples from two external indications: breast cancer (n = 90) and ovarian cancer (n = 50). HistoPLUS maintained robust detection and segmentation capabilities across both tissue types, achieving DQ scores of 0.836 [0.819–0.852] for breast and 0.805 [0.782–0.825] for ovarian samples, with corresponding SQ scores of 0.801 [0.796–0.807] and 0.803 [0.795–0.810], respectively (Table 2; Supplementary Table 6). The model demonstrated particularly strong performance in classifying lymphocytes across both indications, with F1 scores of 0.799 [0.757–0.829] on breast and 0.639 [0.585–0.676] on ovarian cases (Supplementary Table 7). It also displayed promising results on cancer cells, with F1 scores of 0.682 [0.594–0.751] for ovarian and 0.475 [0.302–0.620] for breast cases.

3.4

HistoPLUS enables interpretable biomarker analysis in downstream clinical tasks

To explore the clinical applicability of HistoPLUS, we evaluated whether cell-level features extracted from our model could show preliminary ability to predict clinically relevant biomarkers.

We assessed the retrospective prediction of FGFR3 mutations in BLCA using the TCGA-BLCA cohort. To ensure comparison with prior work, we restricted our analysis to the same 290 cases used by Velmahos et al.[55] We extracted cell type densities, proportions, and morphological features from HistoPLUS outputs and trained a multi-variate logistic regression model using 5-fold cross-validation with three repeats. Our model achieved an AUC of 0.80 [0.77–0.83], outperforming both a clinical baseline (AUC = 0.57 [0.54–0.60]) and the deep learning approach reported by Velmahos et al.[55] (AUC = 0.67). Univariate analysis revealed that global densities of lymphocytes ($p < 0.001$), plasmocytes ($p = 0.0015$), and fibroblasts ($p < 0.001$) were negatively associated with FGFR3 mutations, whereas cancer cell density ($p < 0.001$) and morphological features including mean circularity of cancer cells ($p < 0.001$) and lymphocytes ($p < 0.001$) were

Table 1
Comparison of the distribution of understudied cell populations in our dataset and existing public datasets. Values represent the proportion (%) of each cell type among all annotated nuclei. "/" is indicated if the nuclei type is absent from a given dataset.

Cell types	NuCLS	CoNSeP	Lizard	PanNuke	HistoTRAIN (Ours)
Neutrophils	0.09%	/	0.97%	/	5.88%
Eosinophils	0.01%	/	0.73%	/	1.86%
Macrophages	2.69%	/	/	/	5.29%
Mitotic figures	/	/	/	/	0.21%
Apoptotic bodies	/	/	/	1.53%	5.33%
Endothelial cells	/	/	/	/	2.02%
Smooth muscle cells	/	/	/	/	1.90%

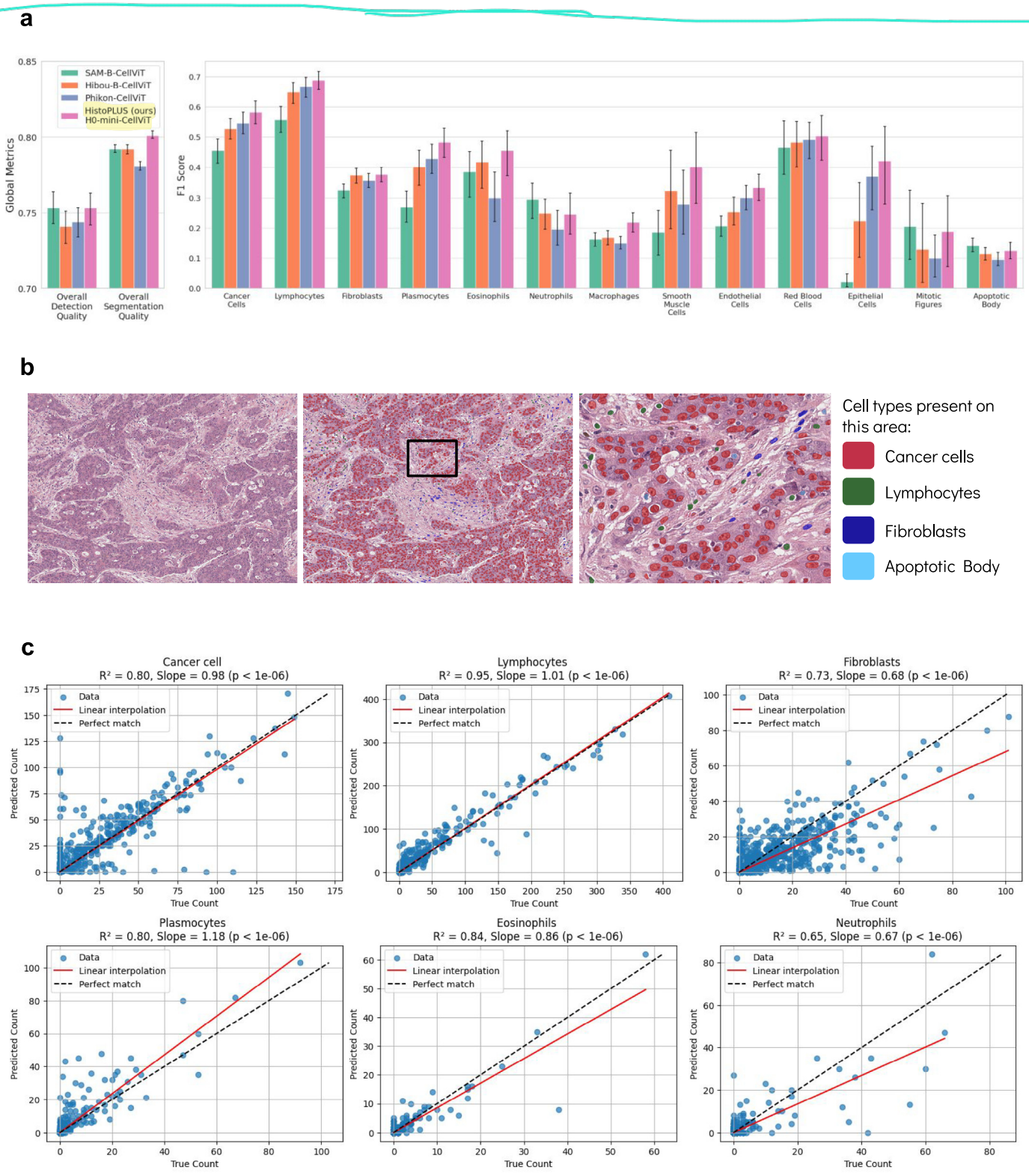


Fig. 2. Inference of HistoPLUS model on unseen slides. (a) Performance of CellViT with vision transformer base encoders on external validation. We evaluated the performance of different vision transformer base encoders integrated with CellViT for cell detection and segmentation in external validation. The grouped barplot on the left presents the median detection, segmentation quality, and F1 scores for each encoder, estimated using 1000 bootstrap iterations. Whiskers indicate the 95% confidence intervals for each metric. (b) Representative H&E image from TCGA-COAD demonstrating nuclear segmentation and cell-type classification. (Left) Original image. (Middle) Annotated region with nuclear boundaries and cell-type labels. (Right) Magnified view of the boxed region in (Middle). (c) Scatter plot showing predicted cell counts (y-axis) vs. ground-truth count (x-axis) in external validation for the six major cell types. Each point represents a tile from HistoVAL. HistoPLUS detected cell counts strongly correlate with annotated ground truths for the most represented cell types in the dataset.

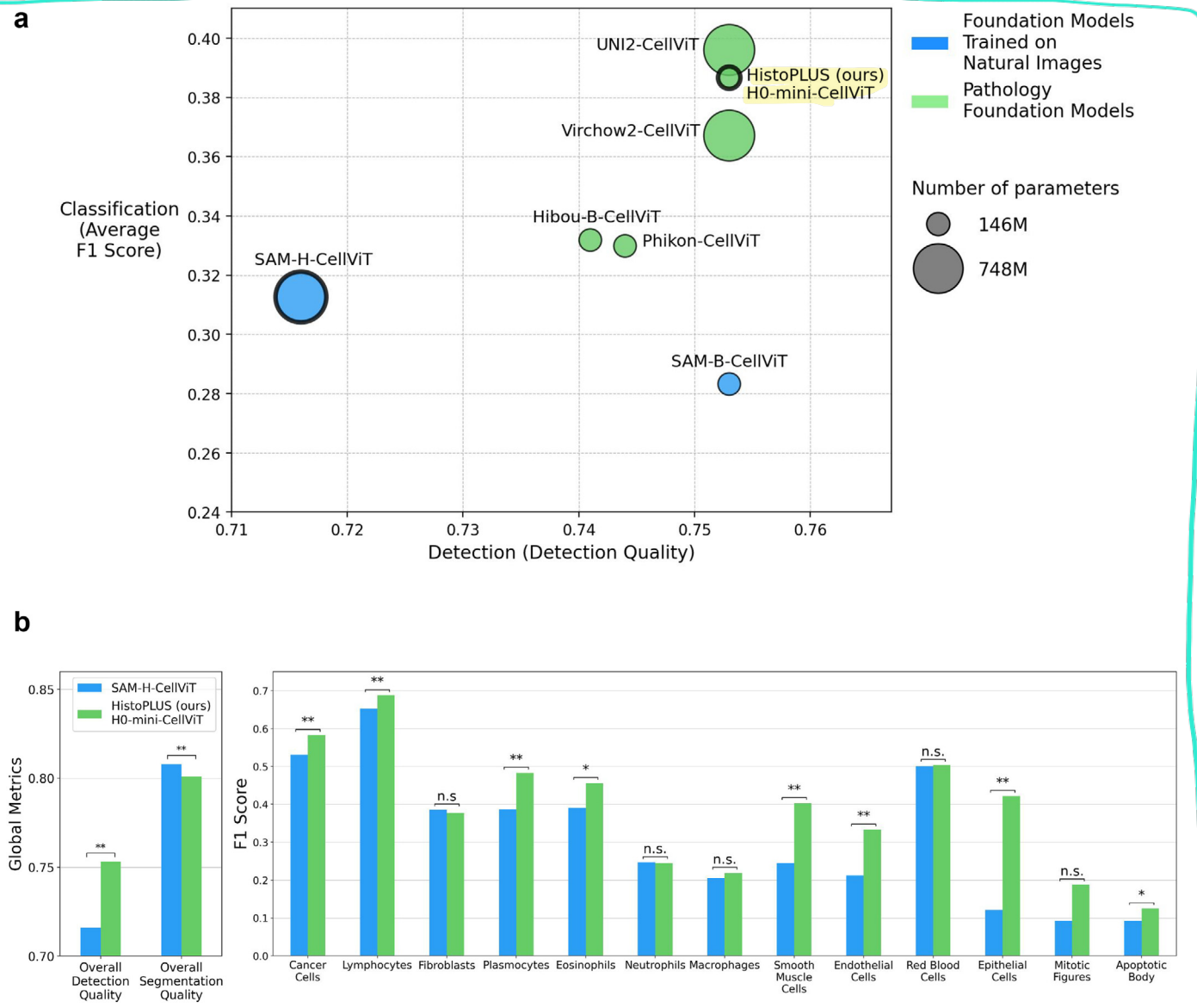


Fig. 3. HistoPLUS outperforms the current state-of-the-art in detection and classification while reducing model complexity. (a) Comparison of CellViT models with different backbones based on detection quality, classification performance, and model size confirm the superiority of pathology-specific encoders on external set. Scatter plot showing detection quality (x-axis) and the average F1 score across cell types (y-axis) for CellViT models with various encoders. Each bubble represents a model, with size indicating the number of parameters. Pathology-specific foundation models are shown in green; general-purpose models in blue. The CellViT model with the H0-mini encoder emerges as the best trade-off, combining strong detection and classification performance with a compact architecture. Models highlighted in bold are those compared in panel b). (b) Head-to-head comparison between CellViT with SAM-H (state-of-the-art) and our model HistoPLUS—CellViT with the H0-mini encoder—on our external test set. The left panel compares detection and segmentation quality, whereas the right panel shows per-class classification F1 scores. Performances are computed via bootstrapping with 1000 iterations. Statistical significance is indicated as follows: $*p < 0.05$, $**p <$ 1e-3; non-significant differences are labeled "n.s.". Double-sided p-values were computed using bootstrap resampling.

Table 2
Zero-shot generalizability of HistoPLUS model on external cohorts. HistoPLUS is validated on different sets of indications: indications that have been seen at training time (validation cohorts for these indications are distinct from cohorts included at training time) and indications that have not been seen at training time. A comprehensive breakdown can be found in Supplementary Tables 8 and 9.

External cohort	Detection quality	Segmentation quality	Cancer cells	Lymphocytes	Fibroblasts	Plasmocytes	Eosinophils	Neutrophils
Reference: External cohorts; seen indications	0.725 [0.713–0.738]	0.801 [0.798–0.804]	0.582 [0.539–0.620]	0.568 [0.534–0.596]	0.339 [0.313–0.366]	0.510 [0.453–0.557]	0.464 [0.373–0.535]	0.255 [0.183–0.327]
Ovarian (external cohort; unseen indication)	0.805 [0.782–0.825]	0.803 [0.795–0.810]	0.682 [0.594–0.751]	0.639 [0.585–0.676]	0.474 [0.417–0.526]	0.330 [0.183–0.430]	0.250 [0.000–0.889]	0.112 [0.000–0.261]
Breast (external cohort; unseen indication)	0.836 [0.819–0.852]	0.801 [0.796–0.807]	0.475 [0.302–0.620]	0.799 [0.757–0.829]	0.463 [0.409–0.523]	0.296 [0.120–0.380]	0.429 [0.100–0.727]	0.139 [0.033–0.245]

positively associated (Fig. 4a). These findings are consistent with prior reports linking FGFR3-altered tumors to reduce immune infiltration[56] and distinct microenvironment profiles.[57]

Discussion

We present HistoPLUS, a model that advances the state-of-the-art of cell detection, segmentation, and classification. Our results demonstrate that pathology-specific foundation models, and diverse training data together significantly increase performance without requiring overly large models, which is crucial for clinical applications.

The performance gains of HistoPLUS reflect two complementary advances. First, our curated training dataset addresses a critical limitation in existing public resources: the under-representation of some clinically important cell types. By employing active learning to systematically enrich our dataset with challenging examples across six cancer types, we achieved more robust classification of infrequent populations such as eosinophils, neutrophils, and endothelial cells. Second, the integration of PFMs as encoders provides substantial improvements over general-purpose vision models, with domain-specific pretraining proving particularly beneficial for rarer cell type classification.

These technical advances open new avenues for H&E-based biomarker discovery beyond traditional density-based and morphometric features. The ability to accurately quantify diverse cell populations at scale enables the development of sophisticated cellular composition signatures and spatial relationships that could complement or potentially replace more expensive molecular assays. For example, precise quantification of tumor-infiltrating immune cell subtypes, including non-lymphoid populations like neutrophils or eosinophils, may provide insights into inflammatory processes and treatment response prediction. Furthermore, when integrated with tissue-level spatial analysis models, HistoPLUS could enable the characterization of cellular microenvironments—such as neutrophil density in juxtatumoral regions or plasmocyte distribution relative to tumor boundaries—potentially revealing spatial biomarkers invisible to current analysis approaches.

Whereas our cross-indication validation demonstrates promising generalizability, several limitations warrant consideration. First, the training dataset was deliberately enriched for under-represented cases, enabling broader coverage of rare cell types but resulting in a distribution that does not fully reflect routine clinical practice. The validation set was largely driven by random selection and should therefore more closely approximate an unbiased clinical sampling distribution (see Supplementary Fig. 3). Nevertheless, further work is needed to fully characterize performance across tissue contexts, specific cancer subtypes, additional cancer types, and non-neoplastic conditions before broader clinical adoption. Additionally, the per-class confusion matrix provides a more granular interpretation of model performance on each cell type and helps understand current model challenges (see Supplementary Fig. 4). Some classes, particularly red blood cells, apoptotic bodies, neutrophils, and macrophages, are sometimes missed, indicating that incomplete detection remains a key limitation. In addition, distinctions are more challenging among morphologically similar stromal populations, such as fibroblasts, endothelial cells, and smooth muscle cells or within smaller structures, including mitotic figures and apoptotic bodies. These findings suggest that the model should be used as an assistive tool with pathologist review and external task-specific validation before clinical deployment. Nevertheless, HistoPLUS represents a significant step toward scalable, comprehensive cellular analysis of routine histopathology, with the potential to uncover new H&E-based biomarkers and more accurate therapeutic stratification.

To illustrate this potential, we conducted an exploratory analysis using HistoPLUS-derived cellular features to predict FGFR3 mutation status in bladder cancer, a clinically actionable alteration that determines eligibility for targeted therapies such as erdafitinib.[58] Whereas molecular testing remains the gold-standard, these exploratory findings suggest that H&E-derived cellular phenotypes may have potential as cost-effective screening tools to prioritize samples for confirmatory testing, or provide complementary predictive information in settings where molecular profiling is unavailable. More broadly, this analysis illustrates how systematic analysis of cellular features by HistoPLUS could support patient characterization and biomarker identification, pending further validation.

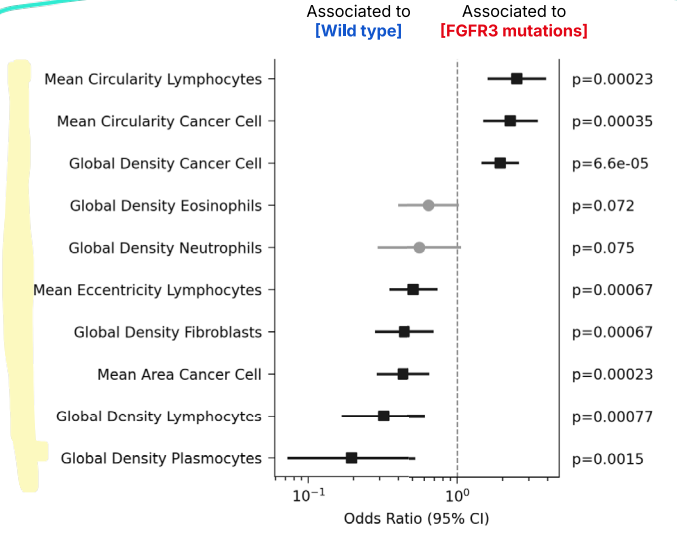


Fig. 4. HistoPLUS supports interpretable biomarker analysis in downstream clinical tasks. (a) Univariate associations between HistoPLUS-derived cellular features and FGFR3 mutation status in bladder urothelial carcinoma. Forest plot showing odds ratios (ORs) with 95% confidence intervals (CIs) for the top 10 histomorphometric covariates ranked by cross-validated AUC. ORs were derived from univariate logistic regression models fitted on standardized feature values (n = 290, TCGA-BLCA). OR > 1 indicates positive association with FGFR3 mutation; OR < 1 indicates negative association. Black squares denote statistically significant associations after Benjamini–Hochberg FDR correction; gray circles denote non-significant associations. The vertical dashed line indicates OR = 1 (null effect). p-values are displayed on the right.

Conclusion

HistoPLUS demonstrates that pathology-specific foundation models combined with strategically curated training data can enable accurate, scalable, and computationally efficient cellular analysis of routine H&E slides. By improving rare cell classification and enabling comprehensive spatial phenotyping, the framework provides a foundation for next-generation H&E-based biomarkers. With further validation and optimization, HistoPLUS has the potential to support biomarker identification, therapeutic stratification, and more accessible precision oncology in routine clinical practice.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpi.2026.100696.

Data availability

TCGA data consists of whole-slide images and can be accessed through the NIH genomic data commons (https://portal.gdc.cancer.gov). The MOSAIC data are not publicly available but access to a subset of 60 patients can be requested through MOSAIC-Window (https://www.mosaic-research.com/mosaic-window). The PanNuke dataset is available at https://huggingface.co/datasets/RationAI/PanNuke and the Lizard dataset at https://conic-challenge.grand-challenge.org/.

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